

Given: $\vec{F}_B = \begin{bmatrix} F_{xB} \\ F_{yB} \\ F_{zB} \end{bmatrix}$, $\vec{M}_B = \begin{bmatrix} M_{xB} \\ M_{yB} \\ M_{zB} \end{bmatrix}$, $\vec{e}_i = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$\phi_i = \theta_i = \psi_i = 0$

Nav. Frame: $\overset{x}{U}, \overset{y}{E}, \overset{z}{N}$

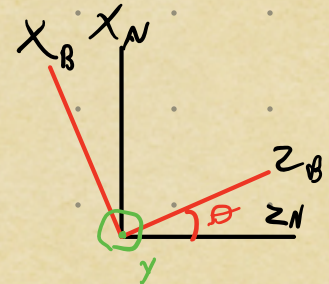
Body Frame: $\overset{x}{x}, \overset{y}{y}, \overset{z}{z}$
 ϕ roll, aligned w/ East, aligned w/ North

Rot. order: ZYX, passive rot.

δ : Gimbal in pitch axis; γ : Gimbal in yaw axis

Find: $\ddot{X}_N, \ddot{Y}_N, \ddot{Z}_N, \ddot{\phi}_{NB}, \ddot{\theta}_{NB}, \ddot{\psi}_{NB}$

$\dot{X}_N, \dot{Y}_N, \dot{Z}_N, \dot{\phi}_{NB}, \dot{\theta}_{NB}, \dot{\psi}_{NB}$



Assume: m is constant; $I_1 = I_2 = I$; model sys. as particle

$$\vec{F}_N = \vec{T}_N - m \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}_N; \quad \vec{T}_N = R_{NB} \vec{T}_B = R_{BN}^T \vec{T}_B$$

$$\vec{T}_B = ||T|| \begin{pmatrix} \cos \delta \cos \gamma \\ \sin \gamma \\ -\sin \delta \end{pmatrix}$$

(BN is passive rot. in neg. dir $\rightarrow$ active rot)

$$\vec{T}_N = \begin{pmatrix} \cos \psi \cos \theta \\ -\cos \psi \sin \theta \sin \phi + \cos \phi \sin \psi \\ \sin \theta \sin \psi + \cos \theta \cos \psi \sin \phi \end{pmatrix} \begin{pmatrix} -\cos \theta \sin \psi \\ \cos \phi \cos \psi + \sin \phi \sin \psi \sin \theta \\ \cos \phi \sin \psi \sin \theta - \cos \psi \sin \phi \end{pmatrix}$$

$$\begin{pmatrix} \sin \theta \\ \cos \theta \sin \phi \\ \cos \theta \cos \phi \end{pmatrix}_{NB} \begin{pmatrix} \cos \delta \cos \gamma \\ \sin \gamma \\ -\sin \delta \end{pmatrix}_B$$

DCM was derived taking the product of the 3 fundamental DCMs in MATLAB.